

AI platform wars

With all the major technology companies offering Machine Learning as a Service, you'd want to know which one's right for you.

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WHAT DOES MACHINE LEARNING AS A SERVICE MEAN?

Machine learning as a service or MLaaS refers to a collection of automated and semi-automated cloud platforms, which provide infrastructure support to accomplish tasks such as model training, data pre-processing, and model evaluation, along with prediction. Using REST APIs, the prediction results from the machine learning service platforms can be merged with the company's existing internal IT infrastructure.

Amazon Web Services, Google Cloud Artificial Intelligence, IBM Watson, and Azure Machine Learning are some of the accessible cloud MLaaS services.

1. AI PLATFORMS

Artificial intelligence platforms enable users for building and training the various machine and deep learning applications and models in a concise time. Since they provide drag-and-drop functionality for efficiently developing algorithms and models, they can be used by anyone with little or no programming experience. In this way, they are similar to various cloud development platforms. By inputting the data through these models, users

can train their models to perform the required tasks. Amazon Sagemaker, Azure Machine Learning Studio, Google Cloud Machine Learning Engine, and IBM Watson Studio are some of MLaaS offerings for AI platforms.

- 1. Amazon Sagemaker:** It was released in November 2017. It enables data scientists and developers to quickly and easily build, train, and deploy machine learning models at all scales – from personal projects to industrial use.
- 2. Azure Machine Learning Studio:** It was released in February 2015. It is a powerful browser-based simple visual drag and drop tool that requires no coding at all. The drag and drop components can be connected in a variety of ways to customize your model. It also has an interface for visualising the data for the project. The data can be uploaded from various sources.
- 3. Google Cloud Machine Learning Engine:** It replaced the Google Prediction API by releasing in March 2016. Google Cloud Machine Learning Engine lets you build machine learning models with ease. These models can train themselves on any data, and irrespective of the size. It also provides a powerful TensorFlow framework that is specially developed for Machine Learning. It is used in many Google products, like Google Photos and Google Cloud Speech. According to experts, the performance and accuracy of Google Cloud ML services in image analysis, particularly in OCR and multi-object detection, are amazing.
- 4. IBM Watson Studio Neural Networks Modeller:** This provides an intuitive drag and drop interface for designing machine learning and neural network models. It speeds up the entire process since

there is no need to write and debug the code for designing the architecture of a model.

NATURAL LANGUAGE PROCESSING

The segment of AI that allows for the interaction between applications and human languages is called Natural language processing (NLP). NLP has many diverse applications like machine translation, sentiment analysis, grammar parsing and part-of-speech tagging. It is also the core technology behind chatbots.

- 1. Amazon Polly:** It is a text-to-speech service using deep learning technologies at its heart for synthesizing speech to make it sound like a human voice. Hence, it allows you to create applications that can talk with a human-like voice, thus creating new speech-enabled products.
- 2. Amazon Lex:** It enables the conversion of speech to text by using the powers of advanced deep learning libraries for automatic speech recognition (ASR). Furthermore, it uses natural language understanding (NLU) for recognizing the tone of the text, enabling the development of applications with lifelike conversational interactions and strongly engaging user experiences.
- 3. Amazon Translate:** It is a neural machine translation service which provides high-quality, and fast language translation by using machine learning and deep learning models for delivering more accurate and more natural sounding translation. This technique delivers better translation than traditional rule-based and statistical translation algorithms.
- 4. Amazon Comprehend NLP:** This natural language processing tool uses the power of machine learning to find insights and relation-